- $\det = ad - bc \rightsquigarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}$
- $\text{traço} = a + d \rightsquigarrow$
- $\lambda^2 - \text{traço}\,\lambda + \det = 0$

↳ $x = \dfrac{-(\text{traço}) \pm \sqrt{(\text{traço})^2 - 4 \cdot a \cdot \det}}{2 \cdot a}$

$(A - \lambda I)V = 0$

Caso de dois valores.
Caso de valor indeterminado.
Caso de valor único.
* Simplificar a Solução geral.
* montar sistema $\begin{cases} x + y = z \\ x - z = y \end{cases}$
* encontrar C_1 e C_2 e substituir

P3

Q1- $x' = \begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix}$

- $\det = 4 \cdot 0 - (-1)3 = 3$
- $\text{traço} = 4 + 0 = 4$
- $\lambda^2 - 4\lambda + 3 = 0$

$x = \dfrac{-(-4) \pm \sqrt{(-4)^2 - 4 \cdot 1 \cdot 3}}{2}$ ↳ 3, 1

Caso de 2 valores

Para $\lambda = 3$:

$(A - 3I)V = 0$

$\begin{pmatrix} 4-3 & -1 \\ 3 & 0-3 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}$ ↳ $x - y = 0$, $3x - 3y = 0$

$V_1 = V_2$ ↝ $x = y$

Auto Vetor $= \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Para $\lambda = 1$

$(A - I)V = 0$

$\begin{pmatrix} 4-1 & -1 \\ 3 & 0-1 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix}$ ↳ $3x - y = 0$, $3x - y = 0$ ↳ $y = 3x$, $V_2 = 3V_1$

Auto Vetor: $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$Sg: x(t) = e^{3t} C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + e^{t} C_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$x(t) = C_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ $\quad x(0) = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$x(0) = C_1 e^{2 \cdot 0} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{0} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$x_0 = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$\begin{cases} C_1 + C_2 = 2 \\ C_1 + 3C_2 = -4 \end{cases}$

$2 - C_2 + 3C_2 = -4$

$2 + 2C_2 = -4$

$C_2 = -6/2 = -3$

$\begin{cases} C_1 = 2 - C_2 \\ C_1 = 2 - (-3) = 5 \end{cases}$

Sg $x(t) = 5e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + (-3) e^{t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

Q2. $x' = \begin{pmatrix} 2 & 2 \\ -1 & 4 \end{pmatrix} x$

- $\det = 2 \cdot 4 - (-1) \cdot 2 = 10$
- $\text{traço} = 2 + 4 = 6$
- $\lambda^2 - 6\lambda + 10 = 0$

$x = \dfrac{-(-6) \pm \sqrt{(-6)^2 - 4 \cdot 1 \cdot 10}}{2}$

$x = \dfrac{6 \pm \sqrt{-4}}{2} \Rightarrow \dfrac{6 \pm 2i}{2} = 3 \pm i$

$\sqrt{-x} = \sqrt{x} \cdot i$

Caso de valor indeterminado

Para $\lambda = 3 + i$

$(A - (3+i)I)V = 0$ ↳ Sinal altera todos

$\begin{pmatrix} 2-(3+i) & 2 \\ -1 & 4-(3+i) \end{pmatrix} \Rightarrow \begin{pmatrix} -1-i & 2 \\ -1 & 1-i \end{pmatrix}$

$\begin{cases} (-1-i)x + 2y = 0 \rightsquigarrow y = \dfrac{1+i}{2} x \\ -x + (1-i)y = 0 \end{cases}$

- Como y depende de x assumimos $x = 2$ para evitar fração.

$y = \dfrac{1+i}{2} \cdot 2 \Rightarrow y = 1 + i = V_2$ ↳ $V_1 = 2$, $V = \begin{pmatrix} 2 \\ 1+i \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} i$

$Sg \cdot x(t) = e^{3t} \left[C_1 \begin{pmatrix} 2\cos t - 0\sin t \\ \cos t - \sin t \end{pmatrix} + C_2 \begin{pmatrix} 0\cos t + \sin t \\ \cos t + \sin t \end{pmatrix} \right]$

$Sg \cdot x(t) = e^{3t} \left[C_1 \begin{pmatrix} 2\cos t \\ \cos t - \sin t \end{pmatrix} + C_2 \begin{pmatrix} \sin t \\ \cos t + \sin t \end{pmatrix} \right]$

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$$x(t) = e^{3t}\left[C_1\begin{pmatrix}2\cos t\\ \cos t - \operatorname{sen} t\end{pmatrix} + C_2\begin{pmatrix}\operatorname{sen} t\\ \cos t + \operatorname{sen} t\end{pmatrix}\right]$$

$$x(t) = e^{3t}\left[\begin{pmatrix}2C_1\cos t\\ C_1\cos t - C_1\operatorname{sen} t\end{pmatrix} + \begin{pmatrix}C_2\operatorname{sen} t\\ C_2\cos t + C_2\operatorname{sen} t\end{pmatrix}\right]$$

1: $-2C_1\cos t + 2C_2\operatorname{sen} t$ # Simplificar

2: $\underbrace{C_1\cos t - C_1\operatorname{sen} t + C_2\cos t + C_2\operatorname{sen} t}$

$(C_1+C_2)\cos t + (-C_1+C_2)\operatorname{sen} t$

$x(0) = \begin{pmatrix}1\\ -3\end{pmatrix}$

$$x(0) = e^{3\cdot 0}\left[C_1\begin{pmatrix}2\cos 0\\ \cos 0 - \operatorname{sen} 0\end{pmatrix} + C_2\begin{pmatrix}\operatorname{sen} 0\\ \cos 0 + \operatorname{sen} 0\end{pmatrix}\right]$$

$\cos 0 = 1$

$\operatorname{sen} 0 = 0$

$$x(0) = C_1\begin{pmatrix}2\\ 1\end{pmatrix} + C_2\begin{pmatrix}0\\ 1\end{pmatrix} = \begin{pmatrix}1\\ -3\end{pmatrix}$$

$$\begin{cases}2C_1 + 0C_2 = 1 \Rightarrow C_1 = \frac{1}{2}\\ C_1 + C_2 = -3 \Rightarrow \frac{1}{2} + C_2 = -3\end{cases}$$

$C_2 = -3 - \frac{1}{2}$

$C_2 = -\frac{6\cdot 1}{2} = \frac{-7}{2}$

Substituir.

$$x_1(t) = e^{3t}\left(2\cdot\tfrac{1}{2}\cdot\cos t + 2\cdot\left(\tfrac{-7}{2}\right)\cdot\operatorname{sen} t\right)$$

$x_1(t) = e^{3t}(\cos t - 7\operatorname{sen} t)$

$$x_2(t) = e^{3t}\left[\left(\tfrac{1}{2} + \left(\tfrac{-7}{2}\right)\right)\cos t + \left(-\tfrac{1}{2} + \left(\tfrac{-7}{2}\right)\right)\operatorname{sen} t\right]$$

$x_2(t) = e^{3t}(-3\cos t - 4\operatorname{sen} t)$

$$x(t) = e^{3t}\begin{pmatrix}\cos t - 7\operatorname{sen} t\\ -3\cos t - 4\operatorname{sen} t\end{pmatrix}$$

Q_3 - $X' = \begin{pmatrix}5 & -4\\ 4 & -3\end{pmatrix}X$

- $\det = 5\cdot(-3) - (-4)\cdot 4 = 1$
- $\operatorname{traço} = 5 + (-3) = 2$
- $\lambda^2 - 2\lambda + 1 = 0$

$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4\cdot 1\cdot 1}}{2}$

$\lambda = 1$

* Caso Vdds Unico

Para $\lambda = 1$

$(A - I)V = 0$

$$\begin{pmatrix}5-1 & -4\\ 4 & -3-1\end{pmatrix} \Rightarrow \begin{pmatrix}4 & -4\\ 4 & -4\end{pmatrix} \Rightarrow 4v_1 - 4v_2 \Rightarrow v_1 = v_2$$

$V = \begin{pmatrix}1\\ 1\end{pmatrix}$

$(A - I)w = V$

$$\begin{cases}4w_1 - 4w_2 = 1\\ 4w_1 - 4w_2 = 1\end{cases}$$

• Supondo $w_1 = 0$

$-4w_2 = 1 \Rightarrow w = \begin{pmatrix}0\\ -\frac{1}{4}\end{pmatrix}$

$w_2 = -\frac{1}{4}$

$$Sg.\,\&(t) = e^t\left[c_1\begin{pmatrix}1\\ 1\end{pmatrix} + c_2\left(t\begin{pmatrix}1\\ 1\end{pmatrix} + \begin{pmatrix}0\\ -\frac{1}{4}\end{pmatrix}\right)\right]$$

* Simplifica:

$$x(t) = e^t\left[C_1\begin{pmatrix}1\\ 1\end{pmatrix} + C_2\left(\begin{pmatrix}t\\ t\end{pmatrix} + \begin{pmatrix}0\\ -\frac{1}{4}\end{pmatrix}\right)\right]$$

$x_1(t) = C_1 + C_2 t$

$x_2(t) = C_1 + C_2 t - \frac{C_2}{4}$

$x(0) = \begin{pmatrix}0\\ 2\end{pmatrix} \rightsquigarrow x_1(0) = e^0(C_1 + C_2\cdot 0) = 0$

$x_1(0) = C_1 = 0$

$x_2(0) = e^0\left(C_1 + C_2 0 - \frac{C_2}{4}\right) = 2$

$x_2(0) = C_1 - \frac{C_1}{4} = 2$

$x_2(0) = 0 - C_2 = 8$

$x_2(0) \Rightarrow C_2 = -8$

$x_1(t) = e^t(0 + (-8)t) = -8te^t$

$x_2(t) = e^t(0 + (-8)t - \frac{(-8)}{4}) = e^t(-8t + 2)$

$S.\ x(t) = e^t\begin{pmatrix}-8te^t\\ -8t + 2\end{pmatrix}$

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